1.

```
hussain@Istiak:~$ echo hello world
hello world
hussain@Istiak:~$
```

2.

```
MAN(1)                                   Manual pager utils                                   MAN(1)

NAME
       man - an interface to the system reference manuals

SYNOPSIS
       man [man options] [[section] page ...] ...
       man -k [apropos options] regexp ...
       man -K [man options] [section] term ...
       man -f [whatis options] page ...
       man -l [man options] file ...
       man -w|-W [man options] page ...

DESCRIPTION
       man is the system's manual pager.  Each page argument given to man is normally the name of a program, utility or function.  The manual page asso-
       ciated  with  each  of  these arguments is then found and displayed.  A section, if provided, will direct man to look only in that section of the
       manual.  The default action is to search in all of the available sections following a pre-defined order (see DEFAULTS),  and  to  show  only  the
       first page found, even if page exists in several sections.

       The table below shows the section numbers of the manual followed by the types of pages they contain.

       1   Executable programs or shell commands
       2   System calls (functions provided by the kernel)
       3   Library calls (functions within program libraries)
       4   Special files (usually found in /dev)
       5   File formats and conventions, e.g. /etc/passwd
       6   Games
       7   Miscellaneous (including macro packages and conventions), e.g. man(7), groff(7), man-pages(7)
       8   System administration commands (usually only for root)
       9   Kernel routines [Non standard]

       A manual page consists of several sections.

       Conventional  section  names  include NAME, SYNOPSIS, CONFIGURATION, DESCRIPTION, OPTIONS, EXIT STATUS, RETURN VALUE, ERRORS, ENVIRONMENT, FILES,
       VERSIONS, STANDARDS, NOTES, BUGS, EXAMPLE, AUTHORS, and SEE ALSO.

       The following conventions apply to the SYNOPSIS section and can be used as a guide in other sections.

       bold text          type exactly as shown.
       italic text        replace with appropriate argument.
 Manual page man(1) line 1 (press h for help or q to quit)
```

3.

```
LS(1)                          User Commands                          LS(1)

NAME
       ls - list directory contents

SYNOPSIS
       ls [OPTION]... [FILE]...

DESCRIPTION
       List  information  about  the  FILEs  (the current directory by de-
       fault).  Sort entries  alphabetically  if  none  of  -cftuvSUX  nor
       --sort is specified.

       Mandatory arguments to long options are mandatory for short options
       too.

       -a, --all
              do not ignore entries starting with .

       -A, --almost-all
              do not list implied . and ..

       --author
              with -l, print the author of each file

       -b, --escape
              print C-style escapes for nongraphic characters

       --block-size=SIZE
              with  -l,  scale  sizes  by  SIZE  when printing them; e.g.,
              '--block-size=M'; see SIZE format below

       -B, --ignore-backups
              do not list implied entries ending with ~

       -c     with -lt: sort by, and show, ctime (time of last  change  of
              file  status  information);  with -l: show ctime and sort by
              name; otherwise: sort by ctime, newest first

       -C     list entries by columns
 Manual page ls(1) line 1 (press h for help or q to quit)
```

```
hussain@Istiak:~$ who
hussain  pts/1        2026-03-06 19:36
hussain@Istiak:~$
```

```
hussain@Istiak:~$ ls
D1  F2.c       cFile    e3.sh  e6.sh   f1.txt.save  f4.txt  f7.txt.save
D2  OS_lab     cFile.c  e4.sh  f1.sh   f2.SH        f6.txt  file1.txt
F2  Shell1.sh  e2.sh    e5.sh  f1.txt  f3.txt       f7.txt  shell1.sh
hussain@Istiak:~$
```

```
hussain@Istiak:~/task1$ nano file.txt
hussain@Istiak:~/task1$ cat file.txt
Hi im from file.txt
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ pwd
/home/hussain/task1
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ mkdir myfolder
hussain@Istiak:~/task1$ cd myfolder
hussain@Istiak:~/task1/myfolder$
```

```
hussain@Istiak:~/task1$ rmdir myfolder
hussain@Istiak:~/task1$ ls
file.txt
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ cp file.txt file2.txt
hussain@Istiak:~/task1$ ls
file.txt  file2.txt
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ mv file2.txt newfile.txt
hussain@Istiak:~/task1$ ls
file.txt  newfile.txt
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ rm file.txt
hussain@Istiak:~/task1$ ls
newfile.txt
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ touch script.sh
hussain@Istiak:~/task1$ chmod 755 script.sh
hussain@Istiak:~/task1$ ls -l script.sh
-rwxr-xr-x 1 hussain hussain 0 Mar  6 19:50 script.sh
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ sudo chown istiak newfile.txt
hussain@Istiak:~/task1$ ls -l newfile.txt
-rw-r--r-- 1 istiak hussain 20 Mar  6 19:49 newfile.txt
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ echo Hello World
Hello World
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ more newfile.txt
Hi im from file.txt
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ less newfile.txt
```

```
Hi im from file.txt
newfile.txt (END)
```

```
hussain@Istiak:~/task1$ date
Fri Mar  6 20:00:17 +06 2026
```

```
hussain@Istiak:~/task1$ time ls
newfile.txt  script.sh

real    0m0.004s
user    0m0.004s
sys     0m0.001s
```

```
hussain@Istiak:~/task1$ ps
    PID TTY          TIME CMD
    308 pts/0    00:00:00 bash
    686 pts/0    00:00:00 ps
hussain@Istiak:~/task1$
```

```
 645  ls
 646  mkdir task1
 647  cd task1
 648  cat file.txt
 649  touch file.txt
 650  cat file.txt
 651  nano file.txt
 652  cat file.txt
 653  ls
 654  pwd
 655  mkdir myfolder
 656  cd myfolder
 657  rmdir myfolder
 658  cd ..
 659  rmdir myfolder
 660  ls
 661  cp file.txt file2.txt
 662  ls
 663  mv file2.txt newfile.txt
 664  ls
 665  rm file.txt
 666  ls
 667  chmod 755 script.sh
 668  touch script.sh
 669  chmod 755 script.sh
 670  ls -l script.sh
 671  sudo useradd istiak
 672  cat /etc/passwd
 673  cut -d: -f1 /etc/passwd
 674  ls
 675  sudo chown istiak newfile.txt
 676  ls -l newfile.txt
 677  echo Hello World
 678  more newfile.txt
 679  less newfile.txt
 680  date
 681  time ls
 682  ps
 683  kill 1234
 684  history
hussain@Istiak:~/task1$
```

4.

```
hussain@Istiak:~/task1$ cat sample.txt
ello Linux
Linux is powerfu
 like programmin
Linux is open sourc
earning awk and se
hussain@Istiak:~/task1$ awk '{print $1}' sample.txt
ello
Linux
like
Linux
earning
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ cat sample.txt
ello Linux
Linux is powerfu
 like programmin
Linux is open sourc
earning awk and se
hussain@Istiak:~/task1$ grep -c "Linux" sample.txt
3
hussain@Istiak:~/task1$
```

```
hussain@Istiak:~/task1$ cat sample.txt
Hello Linux
Linux is powerful
I like programming
Linux is open source
Learning awk and sed
hussain@Istiak:~/task1$ sed -i 's/^.\(.*\).$/\1/' sample.txt
hussain@Istiak:~/task1$ cat sample.txt
ello Linu
inux is powerfu
 like programmin
inux is open sourc
earning awk and se
hussain@Istiak:~/task1$
```

5.

```
hussain@Istiak:~/task1$ touch test.sh
hussain@Istiak:~/task1$ nano test.sh
hussain@Istiak:~/task1$ ./test.sh
-bash: ./test.sh: Permission denied
hussain@Istiak:~/task1$ chmod u+x test.sh
hussain@Istiak:~/task1$ ./test.sh
HELLO WORLD
hussain@Istiak:~/task1$ sh test.sh
HELLO WORLD
hussain@Istiak:~/task1$
```

6.

```
hussain@Istiak:~/task1$ who > myfile1
hussain@Istiak:~/task1$ more myfile1
hussain  pts/1        2026-03-06 19:36
hussain@Istiak:~/task1$
```

7.

```
hussain@Istiak:~/task1$ date; who > myfile2
Fri Mar  6 22:46:39 +06 2026
hussain@Istiak:~/task1$ more myfile2
hussain  pts/1        2026-03-06 19:36
hussain@Istiak:~/task1$
```

8.

```
hussain@Istiak:~/task1$ touch swap.txt
hussain@Istiak:~/task1$ nano swap.txt
hussain@Istiak:~/task1$ cat swap.txt
Hello World from Linux
I like programming
Swap words now
hussain@Istiak:~/task1$ sed 's/^\([^ ]*\) \([^ ]*\)/\2 \1/' swap.txt
World Hello from Linux
like I programming
words Swap now
hussain@Istiak:~/task1$ sed -i 's/^\([^ ]*\) \([^ ]*\)/\2 \1/' swap.txt
hussain@Istiak:~/task1$ cat swap.txt
World Hello from Linux
like I programming
words Swap now
hussain@Istiak:~/task1$
```

9.

```
hussain@Istiak:~/task1$ more hello.sh

# This script prints HELLO WORLD
echo "HELLO WORLD"
hussain@Istiak:~/task1$ ./hello.sh
HELLO WORLD
hussain@Istiak:~/task1$ more helloWorld.c
#include <stdio.h>

int main() {
    printf("HELLO WORLD from C\n");
    return 0;
}
hussain@Istiak:~/task1$ ./helloWorld
HELLO WORLD from C
hussain@Istiak:~/task1$ time ./hello.sh
HELLO WORLD

real    0m0.010s
user    0m0.005s
sys     0m0.006s
hussain@Istiak:~/task1$ time ./helloWorld
HELLO WORLD from C

real    0m0.011s
user    0m0.000s
sys     0m0.009s
hussain@Istiak:~/task1$
```

10.

```
echo "Enter a file or directory name:"
read name

if [ -d "$name" ]
then
    echo "$name is a directory"

elif [ -f "$name" ]
then
    echo "$name is a regular file"

else
    echo "$name is something else"
fi
```

```
hussain@Istiak:~/task1$ ls
check.sh  hello.sh    helloWorld.c  newfile.txt  swap.txt
d1        helloNew    myfile1       sample.txt   test.sh
hello     helloWorld  myfile2       script.sh
hussain@Istiak:~/task1$ ./check.sh
Enter a file or directory name:
d1
d1 is a directory
hussain@Istiak:~/task1$ ./check.sh
Enter a file or directory name:
test.sh
test.sh is a regular file
hussain@Istiak:~/task1$ ./check.sh
Enter a file or directory name:
hi
hi is something else
hussain@Istiak:~/task1$
```

11.

```
hussain@Istiak:~/task1$ more shell.sh


echo "Enter file names:"
read -a files

for f in "${files[@]}"
do
    if [ -f "$f" ]; then
        tr '[:lower:]' '[:upper:]' < "$f" > temp
        mv temp "$f"
        echo "$f converted to UPPERCASE"
    else
        echo "$f not found"
    fi
done
hussain@Istiak:~/task1$ more upercase.txt
i love bangladesh
hussain@Istiak:~/task1$ ./shell.sh
Enter file names:
upercase.txt
upercase.txt converted to UPPERCASE
hussain@Istiak:~/task1$ more upercase.txt
I LOVE BANGLADESH
hussain@Istiak:~/task1$
```

12.

```
hussain@Istiak:~/task1$ more displays.txt
I love Bangladesh
I am a boy
I want to learn linux
I want to become a software engineer
My dream is to make happy my parents
I love my parents
hussain@Istiak:~/task1$ more displays.sh
#write a shell scripts that accepts
#a file name, starting and ending line number as arguments and
#displays all the line between the given line

echo "Enter file name:"
read filename

echo "Starting line:"
read s

echo "Ending line:"
read e

sed -n "${s},${e}p" "$filename"
hussain@Istiak:~/task1$ ./displays.sh
Enter file name:
displays.txt
Starting line:
2
Ending line:
5
I am a boy
I want to learn linux
I want to become a software engineer
My dream is to make happy my parents
hussain@Istiak:~/task1$
```

13.

```
hussain@Istiak:~/task1$ more delete.sh

echo "enter file name:"
read filename

echo "word containing line want to delete:"
read word

sed -i "/$word/d" "$filename"
hussain@Istiak:~/task1$ more delete.txt
hello world
linux system
good
hussain@Istiak:~/task1$ ./delete.sh
enter file name:
delete.txt
word containing line want to delete:
world
hussain@Istiak:~/task1$ more delete.txt
linux system
good
hussain@Istiak:~/task1$
```

14.

```
hussain@Istiak:~/task1$ more substring.sh
read -p "Enter a string: " str
read -p "Enter start character: " st
read -p "Enter substring length: " si

pos=$(expr index "$str" "$st")
substring=${str:pos-1:si}

echo "Substring: $substring"
hussain@Istiak:~/task1$ ./substring.sh
Enter a string: I Love Bangladesh
Enter start character: o
Enter substring length: 6
Substring: ove Ba
hussain@Istiak:~/task1$
```

15.

```
hussain@Istiak:~/task1$ more lengthofstring.sh
read -p "Enter a string: " str
echo "String length: ${#str}"
hussain@Istiak:~/task1$ ./lengthofstring.sh
Enter a string: I love Bangladesh
String length: 17
hussain@Istiak:~/task1$
```